

CTA-Based Plaque-RADS to Assess Etiologic Role of Nonstenotic Carotid Plaques in Patients With Embolic Stroke of Undetermined Source

Angelo Cascio Rizzo,¹ Ghil Schwarz,¹ Simone Bellavia,¹ Andrea Magi,² Guglielmo C. Pero,³ Antonio Macera,³ Amedeo Cervo,³ Francesca Poggetti,² Antonella Moreo,⁴ Valerio Tolva,⁵ Elio Clemente Agostoni,¹ Mariangela Piano,³ and Maria Sessa¹

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Correspondence

Dr. Cascio Rizzo
angelo.casciorizzo@
ospedaleniguarda.it

Abstract

Background and Objectives

Nonstenotic (<50%) carotid plaques (NSCPs) are increasingly recognized as potential embolic sources in embolic stroke of undetermined source (ESUS), but their etiologic role remains uncertain because of limited data and the lack of standardized definition of high-risk plaque. We evaluated the clinical applicability and reliability of a modified CT angiography (CTA)-based Plaque-Reporting and Data System (RADS) classification for characterizing NSCPs in ESUS and examined the association between high-risk NSCPs and stroke laterality, the coexistence of competing stroke mechanisms, and the risk of stroke recurrence.

Methods

We conducted a single-center cohort study (January 2016–June 2025) including consecutive ESUS patients with unilateral anterior circulation stroke and neck CTA, retrospectively and prospectively enrolled. NSCPs were classified using a modified CTA-based Plaque-RADS: (1) normal vessel wall; (2) maximum wall thickness (MWT) <3 mm; (3) MWT ≥3 mm (3ab) or ulceration (3c); (4) intraplaque hemorrhage (4a) or intraluminal thrombus (4c). Patients were stratified according to the presence of ipsilateral high-risk Plaque-RADS (categories 3–4) vs ipsilateral low-risk Plaque-RADS (categories 1–2). Inter-rater agreement for CTA-based Plaque-RADS was assessed, and associations between high-risk Plaque-RADS and stroke side, competing embolic sources, and recurrent ischemic events were evaluated with multivariable regression and survival analyses.

Results

Among 512 patients (median age 70 years, 47.8% women), 109 (21.3%) had ipsilateral high-risk and 403 (78.7%) had low-risk Plaque-RADS scores. Inter-rater agreement was excellent (Cohen κ = 0.83, p < 0.001). High-risk Plaque-RADS scores were more frequent ipsilaterally than contralaterally (21.3% vs 11.9%; adjusted odds ratio [aOR] 2.65, 95% CI 1.73–4.06, p < 0.001). Ipsilateral high-risk Plaque-RADS scores were inversely associated with cardioembolic sources, including atrial cardiopathy (34.9% vs 39.2%; aOR 0.59, 95% CI 0.36–0.96, p = 0.036), high-risk patent foramen ovale (2.7% vs 21.8%; aOR 0.14, 95% CI 0.03–0.60, p = 0.008), and atrial fibrillation detected after stroke (8.4% vs 17.5%; adjusted subdistribution hazard ratio [aSHR] 0.46, 95% CI 0.23–0.95, p = 0.037). During follow-up, overall stroke recurrence did not significantly differ, but ipsilateral high-risk Plaque-RADS scores were significantly associated with an increased risk of ipsilateral stroke recurrence (11.2% vs 3.0%; aSHR 3.19, 95% CI 1.15–8.83, p = 0.026).

Discussion

CTA-based Plaque-RADS can be reliably applied in routine clinical practice to categorize NSCPs in ESUS. Ipsilateral high-risk NSCPs are strongly associated with the index stroke side, lower prevalence of competing cardioembolic sources, and increased ipsilateral recurrence,

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¹Department of Neurology and Stroke Unit, ASST Grande Ospedale Metropolitano Niguarda, Milan, Italy; ²School of Medicine and Surgery, University of Milano-Bicocca, Milan, Italy; ³Department of Neuroradiology, ASST Grande Ospedale Metropolitano Niguarda, Milan, Italy; ⁴Department of Cardiology, De Gasperi Cardio Center, ASST Grande Ospedale Metropolitano Niguarda, Milan, Italy; and ⁵Vascular Surgery Unit, ASST Grande Ospedale Metropolitano Niguarda, Milan, Italy.

Glossary

AC = atrial cardiopathy; AFDAS = atrial fibrillation detected after stroke; AIS = acute ischemic stroke; aOR = adjusted OR; aSHR = adjusted SHR; CTA = CT angiography; ESUS = embolic stroke of undetermined source; HU = Hounsfield units; IPH = intraplaque hemorrhage; IQR = interquartile range; LAVI = left atrial volume index; LVD = left ventricular disease; LVWMAs = left ventricular wall motion abnormalities; MWT = maximum wall thickness; NSCP = nonstenotic carotid plaque; OR = odds ratio; PFO = patent foramen ovale; RADS = Reporting and Data System; SHR = subdistribution hazard ratio; TOAST = Trial of Org 10172 in Acute Stroke Treatment.

supporting their role as a distinct causal mechanism within ESUS and underscoring the need for randomized trials on secondary prevention.

Introduction

Embolic stroke of undetermined source (ESUS)¹ accounts for up to 20% of ischemic strokes and encompasses a heterogeneous range of potential etiologies. While most of these cases are believed to arise from cardioembolic sources, a significant proportion may result from an artery-to-artery embolism.² This mechanism is attributed to atherosclerotic plaques involving the carotid, vertebrobasilar, intracranial arteries and the aortic arch, which are often underrecognized or underestimated in the absence of significant stenosis.³ The term “nonstenotic carotid plaque” (NSCP) has been used to refer to carotid plaques causing less than 50% luminal narrowing, below the conventional threshold of stroke risk.^{4,5} Despite their mild degree of stenosis, these plaques may exhibit high-risk features, such as a lipid-rich necrotic core, intraplaque hemorrhage (IPH), ulceration, or intraluminal thrombus formation after plaque rupture,⁶ which enhance their embolic potential and make them a plausible, yet often overlooked, cause of stroke. Previous studies have demonstrated that NSCPs are more prevalent in ESUS than in cardioembolic strokes,⁷⁻⁹ are more frequently ipsilateral to the ischemic event,¹⁰ and are more likely to exhibit high-risk features when ipsilateral.^{11,12} Nevertheless, patients with ESUS may also present additional potential embolic sources that could serve as competing etiologic mechanisms, making it difficult to establish a definitive causal link between NSCPs and the index stroke. This overlap complicates accurate etiologic classification and, consequently, the choice of the most appropriate secondary prevention strategy. In addition, considerable heterogeneity persists across studies, particularly regarding the definition of complicated plaque and the identification of high-risk features of plaque vulnerability, thereby limiting the generalizability of findings.¹³ To address this gap, a novel imaging-based carotid classification system—the Plaque-Reporting and Data System (RADS) score¹⁴—has been recently proposed to standardize the assessment of carotid plaque composition and morphology using multiple imaging modalities, including ultrasound, CT, and MRI. The Plaque-RADS score stratifies carotid plaques into 4 categories of severity based on specific imaging features: categories 1 and 2 are considered low-risk plaques, while categories 3 and 4 are

classified as high-risk plaques. Given the widespread use of CT angiography (CTA) in the acute stroke setting, a modified CTA-based version of Plaque-RADS was recently tested in a small ESUS cohort, showing a higher prevalence of high-risk nonstenotic Plaque-RADS subtypes ipsilateral to the qualifying stroke.¹⁵ Building on these preliminary findings, it is crucial to validate this classification in a much larger cohort, not only to assess its large-scale applicability but also to explore its role in relation to other potential embolic sources. Indeed, considering the marked heterogeneity of the ESUS definition—which is increasingly recognized as a provisional construct likely to be refined into distinct etiologic subcategories—Plaque-RADS may represent a valuable tool to stratify patients with ESUS and better identify the subgroup in whom NSCPs may play a causal role. In this context, our first objective was to assess the clinical applicability of a modified CTA-based Plaque-RADS score in a large cohort of ESUS patients with anterior circulation strokes, to determine whether this classification system can be reliably integrated into routine clinical practice. Furthermore, we aimed to explore the clinical and etiologic relevance of NSCPs classified by Plaque-RADS, examining their relationship with the index ischemic event through 3 approaches: (1) analyzing the prevalence of high-risk NSCPs (Plaque-RADS 3 and 4) ipsilateral to the infarct, (2) evaluating the coexistence of potential competing etiologic mechanisms, and (3) assessing the risk of recurrent cerebrovascular ischemic events occurring ipsilateral to the identified carotid plaque.

Methods

Study Protocol and Population

This single-center study was conducted at the Stroke Unit of Niguarda Hospital (Milan, Italy) and included all consecutive patients hospitalized between January 2016 and June 2025, who were discharged with a diagnosis of ESUS, had a unilateral anterior circulation stroke, and underwent neck CTA during hospitalization for the index event. Data collection was retrospective for patients admitted until December 2024, and prospective thereafter. To obtain the final study population, we first identified all consecutive patients admitted with acute

ischemic stroke (AIS) during the study period. At hospital discharge, stroke etiology was initially classified according to the Trial of Org 10172 in Acute Stroke Treatment (TOAST) criteria by the treating stroke neurologist, based on a standardized diagnostic workup including brain imaging (CT and/or MRI), cervical and intracranial vessel imaging, trans-thoracic echocardiography, ECG, and in-hospital cardiac monitoring (additional investigations were performed at the discretion of the treating physician when clinically indicated). For the purpose of this study, all TOAST classifications were independently reviewed by 2 neurologists (A.C.R. and G.S.) who had access to the full medical records, including brain and vascular imaging, cardiac investigations, and laboratory data. They re-adjudicated stroke etiology to ensure consistent classification across all categories; in particular, among patients classified as having stroke of undetermined etiology according to TOAST, those meeting ESUS criteria were included. Discrepancies were resolved by consensus. ESUS was defined according to standard criteria¹ as a nonlacunar stroke in the absence of extracranial or intracranial atherosclerosis causing $\geq 50\%$ luminal stenosis of the artery supplying the ischemic territory, no major cardioembolic sources, and no other identifiable stroke causes. Among patients fulfilling ESUS criteria, we included only those with unilateral anterior circulation stroke and available neck CTA. Patients with posterior circulation stroke, multiterritory stroke (i.e., combined anterior and posterior circulation involvement or bilateral anterior circulation stroke), ipsilateral carotid web, or pregnancy-related stroke were excluded. Within the ESUS population, the following potential cardioembolic sources were systematically assessed: atrial cardiopathy (AC), left ventricular disease (LVD), and high-risk patent foramen ovale (PFO). AC was defined as left atrial enlargement, measured by left atrial volume index (LAVI), based on standard criteria: LAVI > 34 mL/m².¹⁶ LVD was defined by the presence of at least one of the following: reduced ejection fraction 30%–49%, LV wall motion abnormalities (LVWMAs; defined as any hypokinetic, akinetic, or dyskinetic segment of the LV wall, not associated with acute or recent [within 4 weeks] myocardial infarction), LV aneurysm, LV enlargement with or without spontaneous echo-contrast, LV noncompaction with deep trabeculations or spongy LV, or cardiomyopathies (e.g., hypertrophic cardiomyopathy, severe restrictive cardiomyopathy, and cardiac amyloidosis).¹⁷ High-risk PFO was defined based on anatomical (large shunt and/or atrial septal aneurysm) and/or clinical (Risk of Paradoxical Embolism score ≥ 7) features in patients aged < 60 years and on anatomical features alone in those aged ≥ 60 years. PFO screening was conducted in most patients aged < 60 years and selectively in those aged ≥ 60 years (i.e., absence of risk factors or concomitant deep vein thrombosis/pulmonary embolism), by transcranial Doppler at rest and during provocative maneuvers with IV agitated saline injection, and/or by transesophageal echocardiography. As an additional potential etiologic factor, the presence of active cancer was also recorded. Active cancer was defined as a diagnosis of solid or hematologic malignancy within the previous 6 months, ongoing cancer treatment, or

active hematologic disease not in remission.¹⁸ Patients were excluded if cancer-related hypercoagulability was suspected, as suggested by elevated D-dimer levels and/or concomitant arterial or venous thrombosis, in which case they were classified as cancer-associated stroke.

Carotid Imaging and Plaque-RADS Classification

All patients included in the study underwent a standardized multimodal head and neck CT protocol, including non-contrast CT, CTA, and CT perfusion. At our hospital, CTA is the first-line vascular imaging modality in the acute stroke pathway and is routinely performed unless contraindicated (e.g., iodinated contrast allergy or renal impairment). Other modalities such as carotid ultrasound or MRI are used only when CTA is not feasible or nondiagnostic, in subacute strokes, or were mainly used during the earliest years of the study period, when CTA was less systematically incorporated in our acute stroke protocol. All CT examinations were performed with a 64-slice scanner (Siemens SOMATOM Definition Edge), using a standard multiphase CTA acquisition protocol (1 phase from the aortic arch to the vertex, followed by 2 phases covering the whole head; slice thickness 0.6 mm, with MIP reconstructions of varying thickness in axial, coronal, and sagittal planes) throughout the entire study period. CTA and CTP were obtained during the injection of 40 mL of iodinated contrast agent followed by 40 mL of saline flush, at a rate of 5 mL/s. Both original and reconstructed images were stored in the hospital Picture Archiving and Communication System and made available for clinical and research purposes. CTA of the intracranial and extracranial arteries was independently reviewed by 2 blinded neuroradiologists (A.M. and A.C.). Anatomical variants, such as a fetal-type posterior cerebral artery or a persistent trigeminal artery, were specifically assessed. The presence of NSCPs was defined according to the North American Symptomatic Carotid Endarterectomy Trial⁵ criteria. The carotid arteries were assessed from the common carotid artery to the internal carotid artery (up to 2 cm beyond the bifurcation) to evaluate plaque presence, maximum wall thickness (MWT), and features of complicated plaque, including IPH, intraluminal thrombus, and ulceration. Because this study relied exclusively on CTA, a modified CTA-based Plaque-RADS classification was applied. Because fibrous cap status is best assessed through MRI vessel wall imaging, we were unable to define certain original subcategories (3a, 3b, and 4b). Thus, the original Plaque-RADS classification was adapted as follows: (1) normal vessel wall; (2) MWT < 3 mm; (3) MWT ≥ 3 mm (3ab) or ulceration (3c), regardless of MWT; and (4) IPH (4a) and intraluminal thrombus (4c), regardless of MWT. Identification of IPH was performed by measuring attenuation levels in Hounsfield units (HU) within the plaque. This was assessed in a region of interest measuring between ≥ 1 and ≤ 2 mm³, using a threshold of < 25 HU on postcontrast acquisitions to define the presence of IPH. This method has previously demonstrated optimal sensitivity and specificity for IPH detection, effectively distinguishing it from the lipid necrotic core in a histopathologic-

radiologic correlation study.¹⁹ Furthermore, a similar CTA-based Plaque-RADS approach has recently been evaluated in a relatively large patient cohort, demonstrating excellent inter-reader and intrareader agreement.²⁰ The plaque categorization scheme was applied retrospectively and uniformly to all examinations, using the same criteria and reading procedure regardless of the year of acquisition. Plaque-RADS categories 1 and 2 were classified as low-risk, while categories 3 and 4 were classified as high-risk. Figure 1 illustrates representative examples from our cohort using the modified CTA-based Plaque-RADS classification.

Follow-Up Outcomes

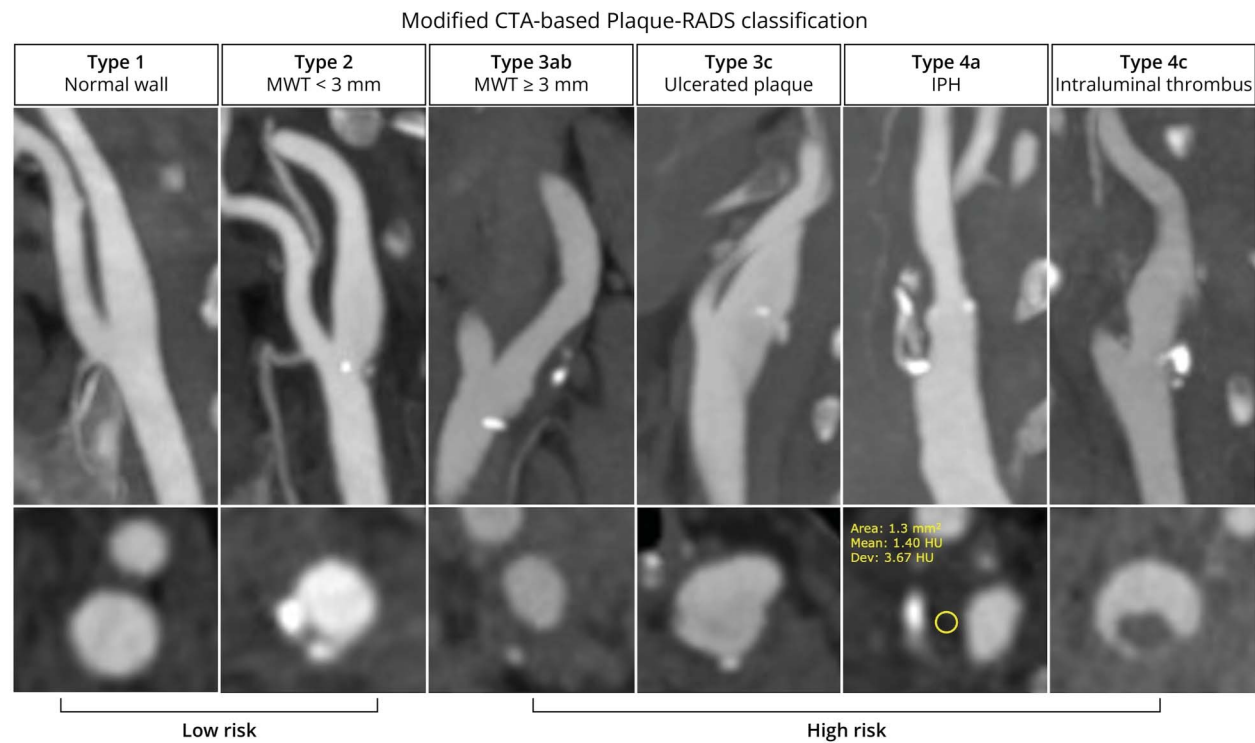
Follow-up data were retrospectively collected from outpatient visits, medical records, and telephone interviews. For patients lost to follow-up, data were censored at the last time known to be alive. Recurrent ischemic cerebrovascular events (ischemic stroke or TIA) were defined as the occurrence of new focal neurologic symptoms after hospital discharge, confirmed by clinical evaluation and, for stroke, by corresponding lesion on brain imaging. Recurrent events were further classified according to laterality relative to the index stroke (ipsilateral vs different vascular territory). Atrial fibrillation detected after stroke (AFDAS) was defined as any newly AF identified after stroke in patients without a history of AF (excluding AF detected during hospitalization).²¹ Outpatient cardiac

monitoring, including Holter monitoring (24 hours–30 days) and/or an implantable loop recorder, was performed at the discretion of the treating physician.

Statistical Analysis

Data were summarized using descriptive statistics: categorical variables as counts and proportions, continuous variables as medians with interquartile ranges (IQRs). The cohort was stratified according to the presence of high-risk plaques ipsilateral to the index stroke (ipsilateral high-risk Plaque-RADS vs ipsilateral low-risk Plaque-RADS). Plaque-RADS classification was independently assessed by 2 blinded neuro-radiologists (A.M. and A.C.), and interrater accuracy was evaluated using the proportion of agreement and Cohen κ statistic. Any disagreement was resolved by a senior neuro-radiologist (M.P.). Differences between ipsilateral high-risk Plaque-RADS vs ipsilateral low-risk Plaque-RADS patients were assessed using bivariate analysis (χ^2 for categorical variables, and the Wilcoxon-Mann-Whitney test for non-normally distributed continuous variables, as appropriate). The mixed-effect logistic regression model (adjusted for age and sex) was used to evaluate the association between high-risk Plaque-RADS and stroke side (ipsilateral vs contralateral). The association between ipsilateral high-risk Plaque-RADS and other potential stroke etiologies (including AC, LVD, PFO, and active cancer) was examined using

Figure 1 Modified CTA-Based Plaque-RADS Classification



Representative examples from our cohort of nonstenotic (<50%) carotid plaques categorized using the modified CTA-based Plaque-RADS classification system. The original Plaque-RADS was adapted as follows: (1) normal vessel wall; (2) MWT <3 mm; (3) MWT ≥3 mm (3ab), and ulceration (3c) regardless of MWT; and (4) IPH (4a) and intraluminal thrombus (4c), regardless of MWT. CTA = CT angiography; HU = Hounsfield units; IPH = intraplaque hemorrhage; MWT = maximum wall thickness; RADS = Reporting and Data System.

multivariable logistic regression to estimate odd ratios (ORs), adjusted for prespecified variables. To investigate whether the presence of ipsilateral high-risk Plaque-RADS was independently associated with the risk of AFDAS and stroke recurrence (overall and ipsilateral), Fine and Gray competing-risk regression was applied, accounting for death as a competing event, to estimate subdistribution hazard ratios (SHRs). Variables associated with the outcome of interest according to current literature were entered into the multivariable models. Multicollinearity was assessed using variation inflation factors. As a sensitivity analysis, cause-specific Cox proportional hazards models were also fitted using the same covariates, treating death as a censoring event. Proportional hazards assumptions were evaluated using Schoenfeld residuals from the cause-specific Cox models, with p value <0.05 considered as evidence of violation. Considering that the causal relationship is quite evident in cases of ipsilateral NSCPs with intraluminal thrombus—and that some classification systems emphasize this as a strong causal association^{22,23}—a sensitivity analysis excluding all patients with ipsilateral Plaque-RADS 4c was performed for all study analyses. Statistical analysis was performed using STATA 17 (Stata Corp. 2021, Stata Statistical Software: Release 17, College Station, TX: Stata Corp LLC). The significance level was set at $p < 0.05$ for 2-sided tests.

Standard Protocol Approvals, Registrations, and Patient Consents

This study was approved by the local Ethical Committee (Comitato Etico Territoriale Lombardia 3, approval number 4998_20.11.2024_N) and was conducted in accordance with the Declaration of Helsinki and with the Strengthening the Reporting of Observational Studies in Epidemiology guideline. For the retrospective portion of the cohort, which relied on data collected as part of routine clinical care and analyzed in anonymized form, the requirement for informed consent was waived. For the prospective cohort (enrolled from December 2024 onward), patients (or their legal representatives) were informed at admission that data collected during routine clinical care could be used for research purposes and provided written informed consent.

Data Availability

Individual deidentified participant data will be made available on reasonable request to qualified investigators, subject to institutional approval and a data sharing agreement.

Results

Study Population

From an initial cohort of 3,105 patients with AIS, 76 (24.8%) were categorized as ESUS according to standard criteria. Of these, 512 met the inclusion criteria (the study flowchart is presented in eFigure 1). Of the 512 patients with ESUS included in the final cohort, 109 (21.3%) had an ipsilateral high-risk Plaque-RADS score and 403 (78.7%) an ipsilateral

low-risk Plaque-RADS score. Baseline characteristics, clinical and cardiac variables, and outcomes are summarized in Table 1. The median age was 70 years (IQR 58–80), the median NIH Stroke Scale score was 7 (IQR 3–13), and 245 patients (47.8%) were women. Ipsilateral high-risk Plaque-RADS patients were older (74 [66–82] vs 69 [56–79] years, $p < 0.001$), were less frequently female (33.0% vs 51.9%, $p < 0.001$), and had a higher prevalence of atherosclerotic cardiovascular risk factors, including hypertension (80.7% vs 69.2%, $p = 0.018$), diabetes (23.8% vs 12.2%, $p = 0.001$), dyslipidemia (66.1% vs 48.6%, $p = 0.005$), and more coronary artery disease (23.8% vs 13.1%, $p = 0.006$). At discharge, ipsilateral high-risk Plaque-RADS patients were more frequently prescribed dual antiplatelet therapy (29.9% vs 11.5%, $p < 0.001$) and high-intensity statins (46.8% vs 23.3%, $p < 0.001$).

Modified CTA-Based Plaque-RADS Inter-Reader Agreement

Among the 1,024 carotid arteries evaluated in 512 patients, 854 (83.4%) were classified as low-risk, including 409 (39.9%) Plaque-RADS 1 and 445 (43.5%) Plaque-RADS 2, whereas 170 (16.6%) were classified as high-risk, including 118 (11.5%) Plaque-RADS 3 (95 [9.3%] type 3ab, 23 [2.2%] type 3c) and 52 (5.0%) Plaque-RADS 4 (36 [3.5%] type 4a, and 16 [1.6%] type 4c) (Table 2). Inter-rater agreement for Plaque-RADS classification was high: the observed agreement was 89.3%, with a Cohen $\kappa = 0.83$ (SE = 0.02; $p < 0.001$), indicating almost perfect agreement.

Prevalence of High-Risk Plaque-RADS Categories and Stroke Side

Of the 170 high-risk plaques, 109 (64.1%) were ipsilateral to the stroke side (21.3% of all ipsilateral carotid arteries), whereas 61 (35.9%) were contralateral (11.9% of all contralateral carotid arteries). The distribution of CTA-based Plaque-RADS categories in all carotid arteries (overall, ipsilateral, and contralateral) is presented in Table 2 and Figure 2. In our mixed-effect logistic regression analysis, the presence of high-risk Plaque-RADS was significantly associated with the ipsilateral side of the stroke (aOR 2.65, 95% CI 1.73–4.06, $p < 0.001$; adjusted for age and sex). The variance of the random intercept was 3.38 (95% CI 1.82–6.30), corresponding to an intraclass correlation coefficient of 0.51 (95% CI 0.36–0.66) (Figure 2, eTable 1).

Coexistence of Competing Stroke Etiologies

In multivariable logistic regression adjusted for prespecified covariates (Table 3), ipsilateral high-risk Plaque-RADS was significantly associated with a lower prevalence of alternative cardioembolic sources. Specifically, patients with ipsilateral high-risk Plaque-RADS showed a reduced prevalence of AC (34.9% vs 39.2%; aOR 0.59, 95% CI 0.36–0.96, $p = 0.036$), high-risk PFO (2.7% vs 21.8%; aOR 0.14, 95% CI 0.03–0.60, $p = 0.008$), and AFDAS (8.4% vs 17.5%; adjusted SHR [aSHR] 0.46, 95% CI 0.23–0.95, $p = 0.037$). No significant differences were observed for LVD (22.0% vs 17.6%; aOR

Table 1 Baseline Characteristics, Clinical and Cardiac Variables, and Outcomes of the ESUS Cohort, Stratified by the Presence of Ipsilateral High-Risk Carotid Plaque-RADS

	Entire cohort (N = 512)	Ipsilateral high-risk Plaque-RADS (N = 109 [21.3%])	Ipsilateral low-risk Plaque-RADS (N = 403 [78.7%])	p Value
Demographics and baseline characteristics				
Age, y, median (IQR)	70 (58–80)	74 (66–82)	69 (56–79)	<0.001
Female sex, n (%)	245 (47.8)	36 (33.0)	209 (51.9)	<0.001
NIHSS score, median (IQR)	7 (3–13)	6 (2–12)	7 (3–14)	0.521
Prestroke mRS score 0–2, n (%)	479 (93.5)	100 (91.7)	370 (94.0)	0.385
Previous stroke, n (%)	49 (9.6)	10 (9.2)	39 (9.7)	0.874
Risk factors and comorbidities				
Smoking (current), n (%)	123 (24.4)	29 (26.6)	94 (23.3)	0.477
Hypertension, n (%)	367 (71.7)	88 (80.7)	279 (69.2)	0.018
Diabetes, n (%)	76 (14.8)	27 (24.8)	49 (12.2)	0.001
Dyslipidemia, n (%)	268 (52.3)	72 (66.1)	196 (48.6)	0.005
LDL-C, median (IQR)	102 (82–126)	102 (83–134)	102 (81–125)	0.523
Obesity, n (%)	90 (17.6)	19 (17.4)	71 (17.6)	0.964
Coronary artery disease, n (%)	79 (15.4)	26 (23.8)	53 (13.1)	0.006
Acute-phase variables, n (%)				
Large vessel occlusion	198 (39.3)	33 (30.3)	165 (40.9)	0.042
IVT	193 (37.7)	40 (36.7)	153 (38.0)	0.809
EVT ± IVT	202 (39.4)	35 (32.1)	167 (41.4)	0.077
None	211 (41.2)	53 (48.6)	158 (39.2)	0.076
Cardiac variables, n (%)				
LAVI, mL/m ² , median (IQR)	31 (25–39)	29 (24–38)	32 (25–39)	0.488
LVEF, %, median (IQR)	59 (56–63)	58 (55–62)	60 (56–63)	0.260
LVWMAs, n (%)	71 (13.9)	16 (14.7)	55 (13.6)	0.782
Implantable loop recorder, n (%)	101 (19.7)	17 (15.6)	84 (20.8)	0.222
Potential embolic sources, n (%)				
Atrial cardiopathy	196 (38.3)	38 (34.9)	158 (39.2)	0.408
Left ventricular disease	95 (18.5)	24 (22.0)	71 (17.6)	0.294
AFDAS	79 (15.5)	9 (8.4)	70 (17.5)	0.022
PFO (high-risk)	91 (17.8) [180]	3 (2.7) [23]	88 (21.8) [157]	<0.001
Active cancer	21 (4.1)	4 (3.7)	17 (4.2)	0.798
Discharge therapy, n (%)				
Antiplatelet therapy	489 (95.5)	107 (98.2)	382 (94.8)	0.131
Dual antiplatelet therapy	76 (15.5) [489]	32 (29.9) [107]	44 (11.5) [382]	<0.001
Anticoagulant therapy	23 (4.5)	3 (2.7)	20 (5.0)	0.323
High-intensity statin therapy	145 (28.3)	51 (46.8)	94 (23.3)	<0.001

Continued

Table 1 Baseline Characteristics, Clinical and Cardiac Variables, and Outcomes of the ESUS Cohort, Stratified by the Presence of Ipsilateral High-Risk Carotid Plaque-RADS (*continued*)

	Entire cohort (N = 512)	Ipsilateral high-risk Plaque-RADS (N = 109 [21.3%])	Ipsilateral low-risk Plaque-RADS (N = 403 [78.7%])	p Value
Outcomes				
90-d mRS score, median (IQR)	1 (0–3) [509]	1 (0–3) [107]	1 (0–3) [402]	0.172
90-d mRS score 0–2, n (%)	368 (72.3)	73 (68.2)	295 (73.4)	0.289
90-d death, n (%)	32 (6.3)	6 (5.7)	26 (6.5)	0.757
Stroke recurrence, n (%)	44 (8.7)	15 (14.0)	29 (7.2)	0.027
Ipsilateral stroke recurrence, n (%)	24 (4.7)	12 (11.2)	12 (3.0)	<0.001

Abbreviations: AFDAS = atrial fibrillation detected after stroke; EVT = endovascular treatment; IQR = interquartile range; IVT = IV thrombolysis; LAVI = left atrial volume index; LDL-C = low-density lipoprotein cholesterol; LVEF = left ventricular ejection fraction; LVWMAs = left ventricular wall motion abnormalities; mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale; PFO = patent foramen ovale; RADS = Reporting and Data System.
 [] Number of patients who underwent the PFO screening test.

0.72, 95% CI 0.36–1.43, $p = 0.349$) or active cancer (3.7% vs 4.2%; aOR 0.77, 95% CI 0.24–2.45, $p = 0.657$).

Risk of Recurrent Cerebrovascular Ischemic Events

Over a median follow-up of 32 months (IQR 14–55), 3 patients (0.6%) were lost at follow-up, 101 patients (19.9%) died, and 44 (8.7%) experienced recurrent stroke (median interval 18 months [IQR 4–40]), of which 24 cases (54.5%) were ipsilateral to the index event. In competing-risk regression (Table 4), accounting for death as a competing risk and adjusted for prespecified variables, no significant difference was observed between ipsilateral high-risk Plaque-RADS and overall stroke recurrence (14.0% vs 7.2%; aSHR 1.48, 95% CI 0.70–3.10, $p = 0.304$); however, ipsilateral high-risk Plaque-RADS was significantly associated with an increased

risk of ipsilateral recurrent ischemic events (11.2% vs 3.0%; aSHR 3.19, 95% CI 1.15–8.83, $p = 0.026$). The median interval for ipsilateral stroke recurrence was 13 months (IQR 4–31), shorter for the ipsilateral high-risk Plaque-RADS group compared with the low-risk Plaque-RADS group (7 [2.5–26] vs 19 [13–36] months). Distribution of ipsilateral recurrent ischemic events (overall and ipsilateral) according to the ipsilateral CTA-based Plaque-RADS subtype is given in eTable 2.

Sensitivity Analysis

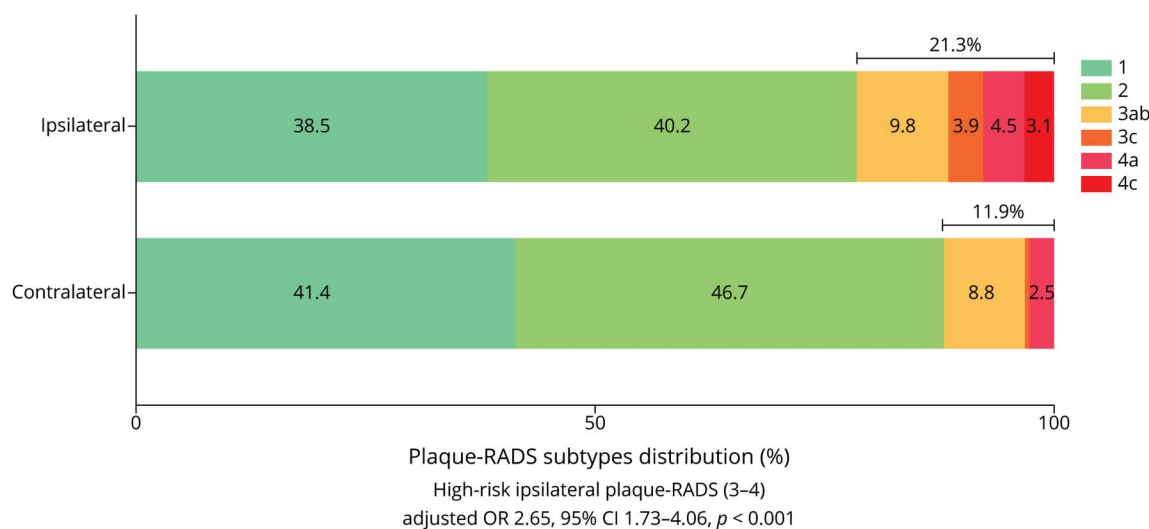
Cause-specific Cox proportional hazards models for time-to-event outcomes (AFDAS and stroke recurrence), which did not account for death as a competing event, yielded results consistent with the competing-risk (Fine-Gray) models (eTable 3). In addition, a sensitivity analysis excluding

Table 2 Distribution of CTA-Based Plaque-RADS Subtypes in Extracranial Carotid Arteries: Overall, Ipsilateral, and Contralateral to the Index Stroke Side

CTA-based Plaque-RADS	All carotid arteries (N = 1,024)	Ipsilateral (N = 512)	Contralateral (N = 512)
Low-risk	854 (83.4)	403 (78.7)	451 (88.1)
1	409 (39.9)	197 (38.5)	212 (41.4)
2	445 (43.5)	206 (40.2)	239 (46.7)
High-risk	170 (16.6)	109 (21.3)	61 (11.9)
3	118 (11.5)	70 (13.7)	48 (9.4)
3ab	95 (9.3)	50 (9.8)	45 (8.8)
3c	23 (2.2)	20 (3.9)	3 (0.6)
4	52 (5.0)	39 (7.6)	13 (2.5)
4a	36 (3.5)	23 (4.5)	13 (2.5)
4c	16 (1.6)	16 (3.1)	0 (0.0)

Abbreviations: CTA = CT angiography; RADS = Reporting and Data System.

Figure 2 Distribution of CTA-Based Plaque-RADS Categories



Distribution of CTA-based Plaque-RADS categories in all carotid arteries, stratified by side relative to the index stroke (ipsilateral vs contralateral), and results of the mixed-effect logistic regression model assessing the association between high-risk Plaque-RADS and stroke side. CTA = CT angiography; RADS = Reporting and Data System.

patients with ipsilateral Plaque-RADS 4c yielded similar results across all study outcomes (sensitivity analyses results are given in eTables 4–7).

Discussion

In this single-center retrospective study, we showed that the application of a modified CTA-based Plaque-RADS classification to categorize NSCPs in ESUS is reliable and reproducible, and we observed a significant association between the presence of high-risk NSCPs and the side of the index stroke. In addition, patients with ipsilateral high-risk NSCPs had a lower prevalence of competing cardioembolic

mechanisms (such as atrial cardiopathy, PFO, or AFDAS) and an increased risk of ipsilateral stroke recurrence, further supporting a potential causal role of the carotid plaque in the ischemic event.

NSCPs are currently considered a minor potential embolic source and are, therefore, included in the ESUS construct. Nevertheless, they represent a clinically relevant subgroup, and a recent update²⁴ has highlighted high-risk NSCPs as a major contributor to ESUS. Despite accumulating data, however, demonstrating a causal relationship between an ipsilateral high-risk NSCP and the index ischemic event remains challenging. Much of the ongoing uncertainty arises from the paucity of large, dedicated studies and the absence of an

Table 3 Regression Analyses for the Association Between Ipsilateral High-Risk Plaque-RADS and Competing Stroke Etiologies

	Univariable analysis		Multivariable analysis	
	OR (95% CI)	p Value	aOR (95% CI) ^a	p Value
Left ventricular disease	1.32 (0.78–2.22)	0.295	0.72 (0.36–1.43)	0.349
Atrial cardiopathy	0.83 (0.53–1.29)	0.408	0.59 (0.36–0.96)	0.036
PFO (high-risk)	0.12 (0.03–0.41)	0.001	0.14 (0.03–0.60)	0.008
Active cancer	0.86 (0.28–2.62)	0.798	0.77 (0.24–2.45)	0.657
	SHR (95% CI)	p Value	aSHR (95% CI) ^b	p Value
AFDAS	0.45 (0.23–0.91)	0.026	0.46 (0.23–0.95)	0.037

Abbreviations: AFDAS = atrial fibrillation detected after stroke; aOR = adjusted OR; aSHR = adjusted SHR; CAD = coronary artery disease; ILR = implantable loop recorder; OR = odds ratio; PFO = patent foramen ovale; RADS = Reporting and Data System; SHR = subdistribution hazard ratio.

Analyses adjusted for prespecified variables:

^a Left ventricular disease, atrial cardiopathy, PFO, and active cancer, adjusted for age, female sex, smoking, hypertension, diabetes, dyslipidemia, and CAD.

^b AFDAS adjusted for age, female sex, smoking, hypertension, diabetes, dyslipidemia, CAD, LVO, atrial cardiopathy, and ILR.

Table 4 Competing-Risk Regression (Fine-Gray) for the Association Between Ipsilateral High-Risk Plaque-RADS and Stroke Recurrence (Overall and Ipsilateral)

	Univariable analysis		Multivariable analysis ^a	
	SHR (95% CI)	p Value	aSHR (95% CI)	p Value
Stroke recurrence	2.06 (1.10–3.86)	0.024	1.48 (0.70–3.10)	0.304
Ipsilateral stroke recurrence	3.91 (1.73–8.83)	0.001	3.19 (1.15–8.83)	0.026

Abbreviations: aSHR = adjusted SHR; CAD = coronary artery disease; DAPT = dual antiplatelet therapy; RADS = Reporting and Data System; SHR = sub-distribution hazard ratio.

^a Adjusted for age, female sex, smoking, hypertension, diabetes, dyslipidemia, CAD, previous stroke, DAPT at discharge, and high-intensity statin at discharge.

operational definition of high-risk NSCPs suitable for multicenter trials. To address this gap, we used the Plaque-RADS classification for the first time in a large cohort of patients with ESUS. This standardized multimodal imaging system was recently designed to harmonize plaque assessment, improve estimation of vulnerability and embolic potential, and enhance reproducibility in both clinical practice and research. Unlike the original version, we used a modified CTA-based version of Plaque-RADS. This choice was motivated by the widespread availability of CTA and routine use in acute stroke evaluation; the ability to classify plaques rapidly, with reduced operator dependence; and—most importantly—its applicability in everyday clinical practice, even in centers without access to advanced imaging techniques. Moreover, this modified CTA-based Plaque-RADS version has recently been evaluated in a relatively large patient cohort, demonstrating excellent inter-reader and intrareader agreement and supporting the reproducibility of this CTA-based approach. Nevertheless, this approach has some limitations. CTA does not allow the assessment of certain plaque features better evaluated with MRI (e.g., fibrous cap integrity and IPH) or ultrasound (e.g., echogenicity and consistency). In particular, the methodology we adopted to define Plaque-RADS 4a subtype—that is, the HU threshold to identify IPH, although previously shown to have optimal sensitivity and specificity for IPH detection—has not yet been validated in large cohorts and carries potential risks of misclassification, as attenuation ranges of plaque components partially overlap on CT—particularly between lipid-rich necrotic core and IPH. However, such limitations may have influenced only the internal subclassifications within categories 3 and 4, which are both considered high-risk, and, therefore, did not affect the overall results of the study. Through this CTA-based approach, we achieved consistent plaque assessment with excellent interobserver agreement, supporting the robustness and transferability of this classification into everyday clinical practice within hospital settings.

In our study, prevalence of high-risk NSCPs was similar to that reported in previous cohorts, with approximately 1 in 5 patients harboring an ipsilateral NSCPs with high-risk embolic features. Moreover, ipsilateral high-risk Plaque-RADS were more prevalent and significantly associated with stroke side compared with contralateral plaques. Consistent with

previous findings, and using the Plaque-RADS classification, our results reinforce the etiologic relevance of high-risk NSCPs in ESUS.¹⁵

Another crucial challenge in attributing causality in ESUS is the frequent overlap of potential embolic mechanisms.²⁵ Indeed, within the ESUS construct, there are several potential etiologic sources, and each could define a specific patient subgroup that, if categorized differently from the broad ESUS category, could benefit from tailored secondary prevention strategies.¹⁵ For example, AC¹⁸ and LVD²⁶ may represent distinct cardioembolic substrates that may require anticoagulation, whereas high-risk NSCPs may instead warrant intensified antiplatelet and lipid-lowering therapies, or interventional strategies. The coexistence of these different embolic sources further complicates etiologic attribution, raising uncertainty about the true causal substrate of the ischemic event and challenging the clinician in selecting the most appropriate therapeutic approach. In our cohort, patients with ipsilateral high-risk Plaque-RADS had a lower prevalence of cardioembolic sources. Specifically, we observed a reduced prevalence of high-risk PFO, supporting previous evidence that NSCPs and PFO may represent independent contributors to ESUS.^{27,28} Interpretation of this finding is limited, however, by the relatively small proportion of patients who underwent systematic PFO screening, most of whom were younger patients with fewer atherosclerotic risk factors. We also found a lower prevalence and inverse association with atrial embolic mechanism (AC and AFDAS), supporting the hypothesis that AF is less frequently detected in ESUS patients with NSCPs compared with those without, although previous studies have reported conflicting results.^{29,30} An additional finding was the absence of any significant association with LVD, which was present in approximately 20% of patients with ipsilateral high-risk Plaque-RADS. LVD (mainly including mild-moderate LV systolic dysfunction, wall motion abnormalities) may itself represent a potential embolic source, for example, through intermittent intraventricular thrombus formation. This overlap may be explained by the similar prevalence of atherosclerotic risk factors that contribute to both carotid and coronary artery disease.

A further approach to explore causality between ESUS and high-risk NSCPs is to evaluate the risk of recurrent

cerebrovascular events. Previous studies showed that plaque features beyond stenosis severity, such as plaque surface irregularity, are strongly associated with an increased risk of ipsilateral stroke,³¹ particularly when high-risk plaque features are present.³² A small population-based study also suggested an increased risk of recurrent ipsilateral events in patients with high-risk NSCPs and cryptogenic stroke.³³ In our cohort, overall recurrence did not differ significantly, but the risk of ipsilateral ischemic events was nearly 3-fold higher in patients with high-risk plaques. This finding suggests that, while overall stroke recurrence may be influenced by different embolic mechanisms, high-risk NSCPs carry an intrinsic embolic potential that warrants careful consideration. At present, most available evidence derives from association studies, and substantial uncertainty remains regarding optimal treatment strategy. A subanalysis of NAVIGATE-ESUS trial found no difference in efficacy between rivaroxaban and aspirin for prevention of recurrent stroke in ESUS patients with carotid atherosclerosis.³⁴ Current guidelines^{35,36} do not recommend carotid revascularization in patients with mild (<50%) symptomatic carotid artery stenosis, but intervention may be considered in selected patients with recurrent events despite best medical therapy (Class IIb, Level C),³⁷ after a full neurovascular workup and multidisciplinary team review. Our study has several strengths. First, we applied a modified CTA-based Plaque-RADS classification, previously tested only in a small ESUS cohort, for the first time in a large and well-characterized population. This approach minimized selection bias ensuring homogeneous diagnostic pathways and providing standardized plaque evaluation with high reproducibility and excellent interobserver agreement. Second, by applying the classification in a larger cohort, we were able to better characterize the distribution of Plaque-RADS subtypes. Moreover, the categorization of IPH directly on CTA using HU thresholds—without the need for additional post-processing software for volumetric quantification¹⁵—highlights the feasibility of this modified version in routine hospital practice. Third, having this large cohort allowed us to adopt a comprehensive methodological approach that incorporated the evaluation of competing embolic mechanisms and long-term follow-up for recurrent events. Taken together, these findings provide robust evidence of the strong association between high-risk NSCP and the index ischemic event, indirectly supporting their etiologic role in ESUS. We also acknowledge several limitations. The single-center, retrospective design may limit generalizability. Only patients undergoing CTA were included, although excluded cases represented a minority. As discussed above, the exclusive use of CTA did not reflect the original classification, which incorporates plaque features better assessed with MRI or ultrasound; in addition, the HU threshold used to define IPH has not yet been validated in large cohorts. Despite comprehensive cardiac evaluation, some subclinical embolic sources (e.g., paroxysmal AF, minor subtle intracardiac thrombus, occult cancer, and underdetected PFO) may have been missed. In addition, our definition of certain cardioembolic sources, such as AC, may be limited, as it does not incorporate

other electrophysiologic or serologic biomarkers. The study spanned a relatively long period (10 years) during which stroke care evolved (e.g., increasing use of dual antiplatelet therapy and changes in diagnostic workup and cardiac rhythm monitoring strategies), which may have introduced heterogeneity in both exposures and potential confounders. Follow-up outcomes were in part assessed retrospectively from medical records and telephone interviews, which may have led to underreporting of recurrent events. Finally, the relatively small number of outcomes precluded subgroup analysis of plaque subtypes, and the observational design cannot establish causality.

Our study demonstrates 2 main advances. First, we showed that a modified CTA-based Plaque-RADS classification can be reliably applied in a large ESUS population, with excellent reproducibility and practical feasibility. This supports its potential integration into routine hospital workflows as a standardized tool for carotid plaque assessment in ESUS. Second, on the etiologic side, we confirmed a strong association between high-risk NSCP and the index ischemic event, reflected in an increased risk of ipsilateral recurrence and a lower prevalence of competing embolic sources. Together, these findings reinforce the notion that high-risk NSCPs may represent a distinct causal mechanism within ESUS and highlight the need for careful clinical attention and comprehensive neurovascular evaluation to optimize secondary prevention. Randomized controlled trials are warranted to assess whether targeted treatments (pharmacologic or interventional) can improve long-term outcomes in this population. Such evidence would help to confirm the causal role of high-risk NSCP, support their rational exclusion from the ESUS construct, and guide individualized preventive strategies to reduce recurrent stroke risk.

Author Contributions

A. Cascio Rizzo: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. G. Schwarz: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. S. Bellavia: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. A. Magi: major role in the acquisition of data. G.C. Pero: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. A. Macera: major role in the acquisition of data; analysis or interpretation of data. A. Cervo: major role in the acquisition of data; analysis or interpretation of data. F. Poggetti: major role in the acquisition of data. A. Moreo: drafting/revision of the manuscript for content, including medical writing for content. V. Tolva: drafting/revision of the manuscript for content, including medical writing for content. E.C. Agostoni: drafting/revision of the manuscript for content, including medical writing for content. M. Piano: analysis or interpretation of data. M. Sessa: drafting/revision of the

manuscript for content, including medical writing for content; analysis or interpretation of data.

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